

What must be measured now to predict after encoding drift?

1 Describe the model

Physically attainable circuit paths

$$a = a(\theta)$$



2 Define the prediction

Same present, different future?

$$Oa = Ob \quad Fa \neq Fb$$

Hidden differences matter
if future responses differ



3 Measure the target

Apply a phase to the unknown target

$$\phi \rightarrow \text{target measurement}$$

Resolve relevant path differences
before the encoding changes



VALIDATE

Equal shot budgets | Blind-phase control | Matched classical model

Forecast gains require persistent hidden structure and a future query sensitive to it.